CANADA'S ARCTIC EMISSION CONTROL AREA (ECA) AND BLACK CARBON EMISSIONS

PRESENTATION TO 2025 POLAR MARITIME SEMINAR

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Environment and Climate Change Canada
Environnement et Changement climatique Canada
Canada

Canada’s Arctic Emission Control Area (ECA)

CONTEXT & RATIONALE

In October 2024, the Canadian Arctic ECA proposal was adopted by the Marine Environment Protection Committee (MEPC 82).

1. The Canadian Arctic was less protected than the rest of North America:

- Marine areas and populations in Canada and the U.S. have benefited from the North American (NA) ECA since it was implemented in 2013.
- The Arctic was not included in the NA ECA due to a lack of modelling capacity in the Arctic and less shipping activity in the region at the time.

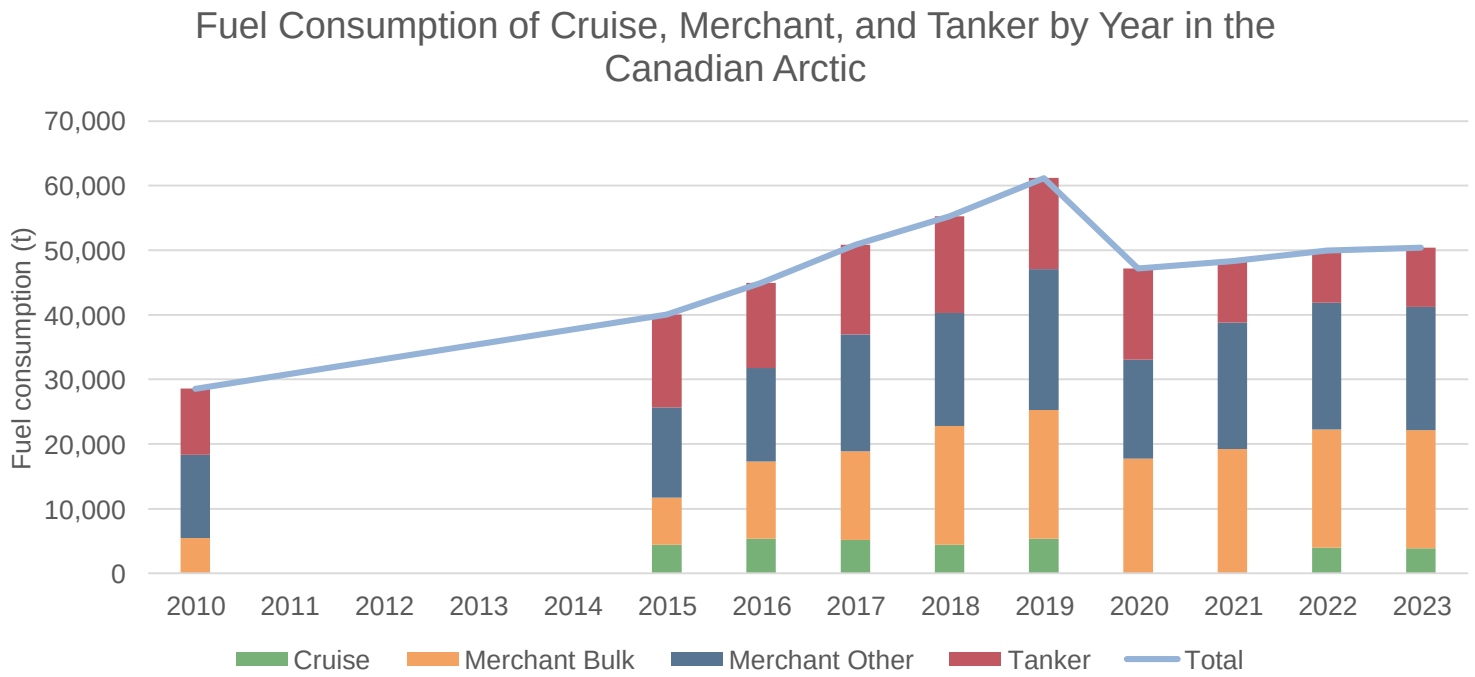
2. Shipping activity and resulting air pollution and climate impacts are increasing in the Canadian Arctic:

- Significant increases in ship traffic have already been observed.
- Increased air pollution greatly impacts health and the environment, the use of cleaner fuels can reduce the warming effects of black carbon.

ARCTIC OCEAN
OCÉAN ARCTIQUE
UNITED STATES OF AMERICA
ÉTATS-UNIS D'AMÉRIQUE
Beaufort Sea
Mer de Beaufort
GREENLAND
GROENLAND
Yukon
Northwest Territories
Nunavut
Ungava Bay
Baie d'Ungava
British Columbia
Hudson Bay
Baie d'Hudson
Newfoundland & Labrador
Alberta
CANADA
Saskatchewan
Manitoba
Québec
James Bay
Baie James
Prince Edward Island
New Brunswick
Ontario
Nova Scotia
PACIFIC OCEAN
OCÉAN PACIFIQUE
ATLANTIC OCEAN
OCÉAN ATLANTIQUE
UNITED STATES OF AMERICA
ÉTATS-UNIS D'AMÉRIQUE
MEXICO
MEXIQUE
Legend / Légende
Proposed Canadian Arctic ECA
Proposition de la ZCE de l'Arctique canadien
North American ECA
ZCE de l'Amérique du Nord
N
0
500
1,000
2,000
Kilometers / kilomètres

INCREASED SHIPPING ACTIVITY

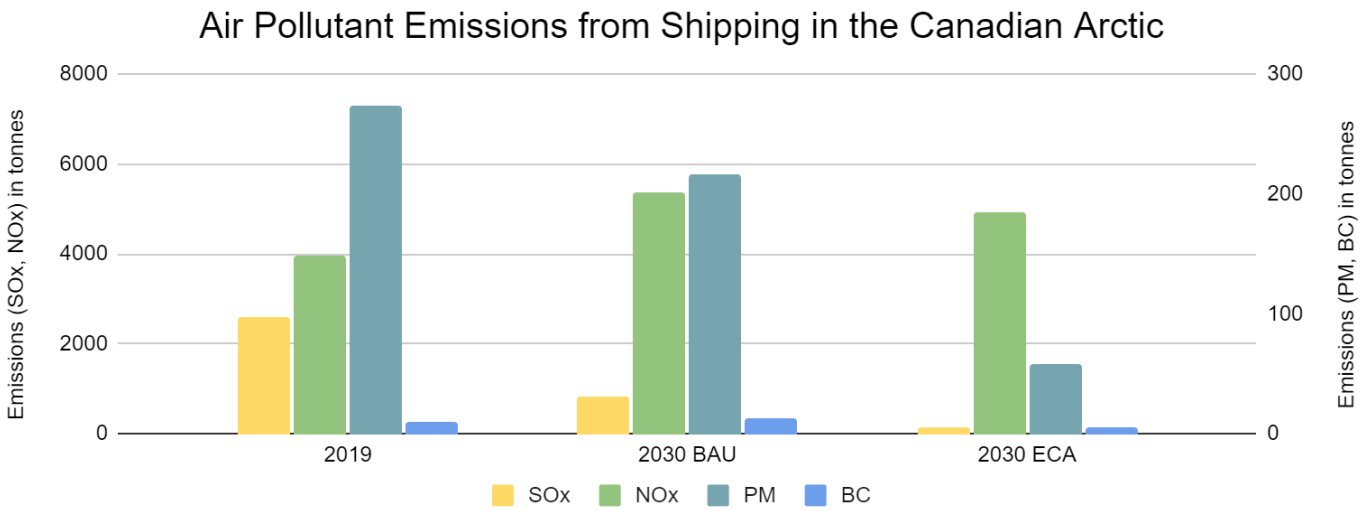
- Between 2010 and 2019, fuel consumption by cruise, merchant, and tanker ships in the Canadian Arctic more than doubled
- COVID-19 caused a decrease in vessel traffic that is still recovering



**An earlier version of the Marine Emissions Inventory Tool was produced for ECCC using 2010 data, ECCC then further developed the tool starting with 2015 data resulting in a gap of data for 4 years*

Ships across this period used various marine fuels, such as HFO, MDO, and VLSFO. It should be noted that the fuel efficiencies of different fuels can differ slightly, which is not accounted for in the graph above

ENVIRONMENTAL BENEFITS



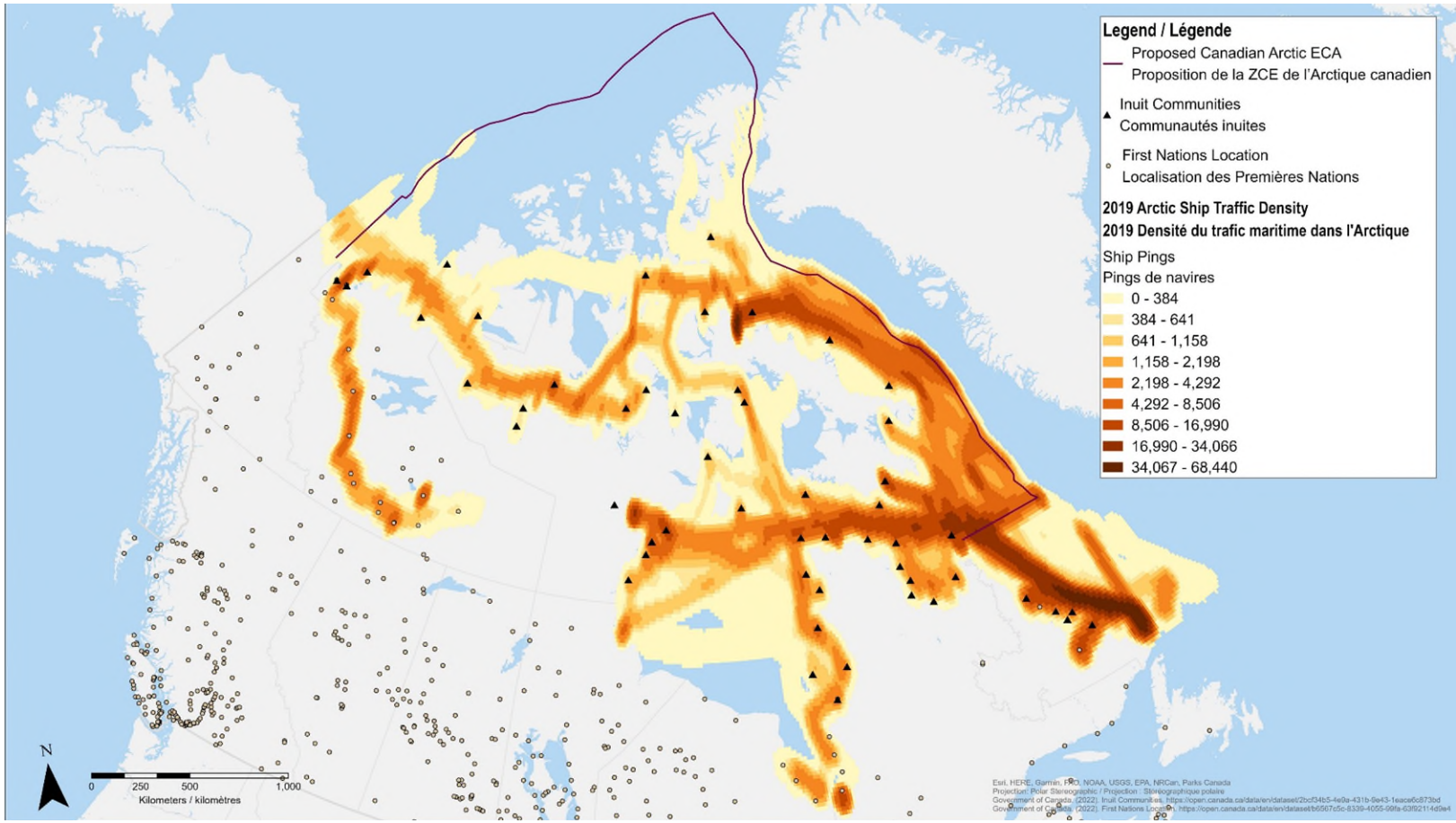
% change from 2030 BAU to ECA	SOx	NOx	Total PM	BC
	-80%	-8%	-74%	-59%

2030 BAU (business-as-usual) scenario: the scenario in 2030 if no additional emissions regulations have been implemented (includes global S cap)
2030 ECA scenario: the scenario in 2030 if ECA and HFO ban regulations have been implemented

- Reduced effects of air pollution, acidification, and eutrophication in the Arctic
  - Critical loads are already reaching exceedance or are exceeded
- Reduction in warming effects due to black carbon (BC) is anticipated to be a co-benefit of the ECA
  - Compliance with the use of scrubbers would not result in the BC benefits
  - Though many vessels have shifted away from HFO usage, 0.5% sulphur fuels (like VLSFO) can still result in higher BC and PM emissions than using ECA-compliant fuels.
  - Other actions besides the ECA are needed for BC reductions in the Arctic as the ECA is the first step and there are land-based sources as well

HEALTH BENEFITS

- Canada has a duty to ensure a healthy environment for all its citizens as we journey towards reconciliation with Indigenous peoples.
- Lower ship emissions:
  - Will reduce impacts to culturally significant areas, upon which Indigenous communities depend for their food security, cultural identity, and way of life
  - Will help to protect vulnerable populations which can be more sensitive to the adverse health effects of pollution

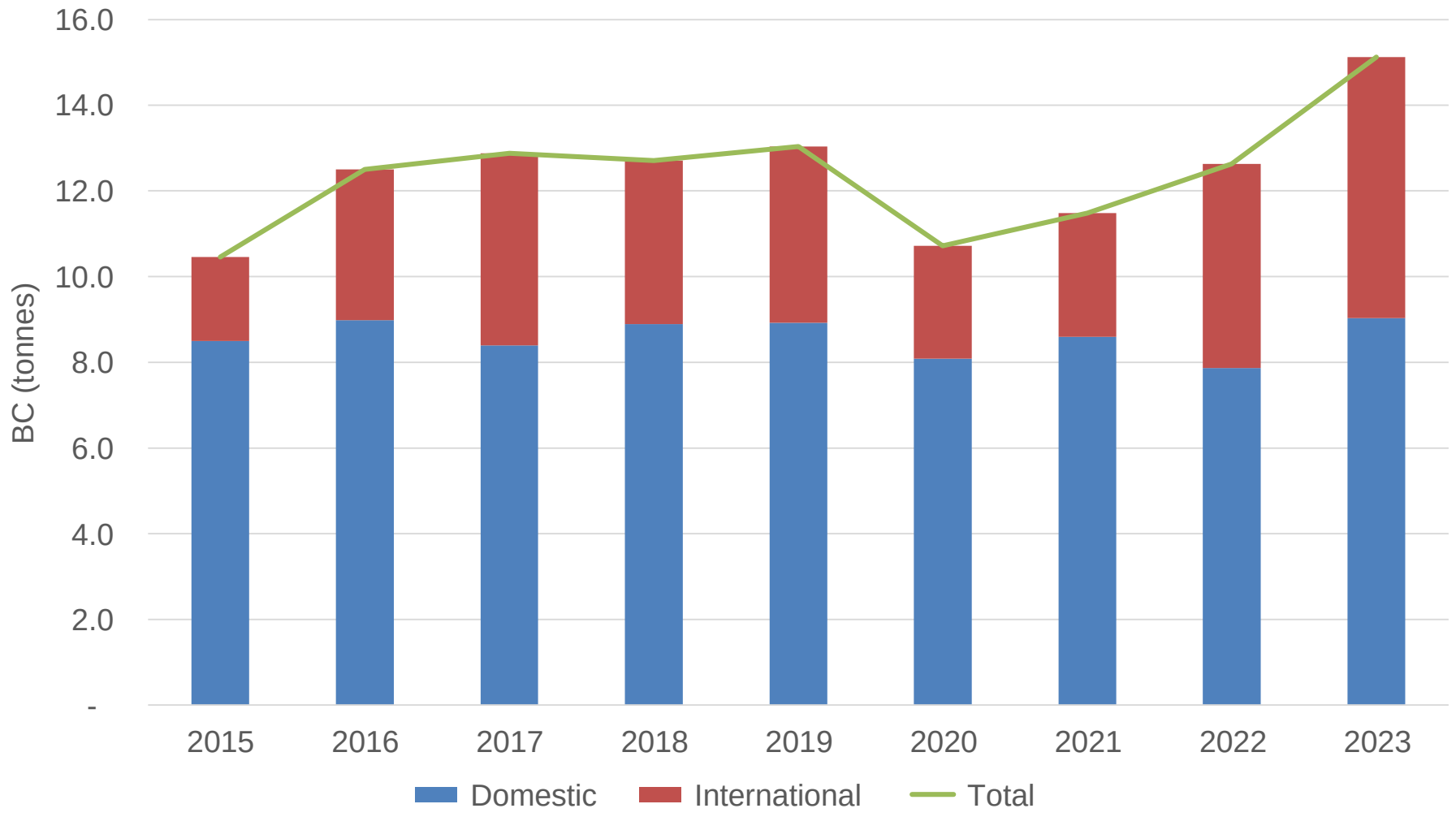


TIMELINE

- **2016-2020**: Arctic research work
- **2021-2023**: Development of Arctic ECA proposal
- **April 2024**: Approval of Arctic ECA proposal at MEPC 81
- **October 2024**: adoption of Arctic ECA proposal at MEPC 82
  - **January 1, 2025**: Ships constructed on or after 1 January 2025 will have to comply with the NOx Tier III standards in the ECA area
  - **September 1, 2025**: Acceptance into MARPOL
  - **March 1, 2026**: Entry into force under MARPOL
  - **March 1, 2027**: SOx regulations domestic entry into force (under MARPOL, all ECA SOx regulations have a one-year delay)

Canada’s Black Carbon Marine Emissions

Domestic and International Black Carbon Shipping Emissions in Canada's Arctic



Source: [Canada's Marine Emissions Inventory Tool]

REDUCING BLACK CARBON EMISSIONS IN THE CANADIAN ARCTIC

- Canada has developed an Arctic and Northern Policy Framework and Strategy that includes emissions reductions commitments
- Marine-based reductions from 0.5% Sulphur Cap, HFO ban, and future Canadian Arctic ECA
- Land-based reductions from programs to invest in clean energy projects, lowering diesel usage and wood-burning:
  - Northern Responsible Energy Approach for Community Heat and Electricity (REACHE) program
  - Clean Energy for Rural and Remote Communities Program
  - Investing in Canada Infrastructure Program
  - Indigenous Off-Diesel Initiative

REDUCING BLACK CARBON EMISSIONS ACROSS CANADA

- Canada is on track to do its part to achieve the collective goal of Arctic States to reduce emissions of black carbon by 25-33% from 2013 levels by 2025
- Canada’s strategies and action plans
  - Strategy on Short-Lived Climate Pollutants
  - 2030 Emission Reduction Plan
  - Clean Fuel Regulations
  - Pan-Canadian Framework on Clean Growth and Climate Change
- Land-based regulations
  - Off-road Compression-Ignition and Large Spark-Ignition Engine Emission Regulations
  - On-Road Vehicle and Engine Emission

THANK YOU

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